=== UNIT 1: INTRODUCTION TO MECHANICS & FORCE SYSTEMS ===

>> 1.1 Foundations of Classical Mechanics and Idealizations

Engineering Mechanics is the foundational physical science that predicts and describes the state of rest or motion of bodies subjected to external force systems. In civil and structural engineering, mechanics governs the design of bridges, multi-story frames, dams, and retaining walls.

* Fundamental Idealizations: (1) Particle: A body of negligible dimensions whose mass is concentrated at a single geometric point (rotational effects neglected). (2) Rigid Body: A continuum of matter wherein relative distances between any two constituent particles remain invariant under applied force systems (deformations neglected in rigid body statics). (3) Concentrated Force: An idealization representing the action of a force applied over an infinitesimal contact surface area.
* Newton’s Laws of Motion: (First Law) A particle remains at rest or moves with constant velocity in a straight line unless acted upon by an unbalanced resultant force. (Second Law) The rate of change of linear momentum is proportional to the resultant force and occurs along its line of action: F = m*a. (Third Law) To every action, there is an equal and opposite reaction: F_AB = - F_BA.
* Principle of Transmissibility: A force acting on a rigid body may be transmitted along its line of action to any other point on the same body without altering the external external effects (external equilibrium). Note: Internal stress distributions depend on the exact application point.

>> 1.2 Parallelogram Law, Triangle Law and Polygon Law of Forces

* Parallelogram Law of Forces: If two concurrent forces acting simultaneously at a common point are represented in magnitude and direction by two adjacent sides of a parallelogram, their resultant is represented by the diagonal passing through their point of intersection.
Formula: R = sqrt(P^2 + Q^2 + 2*P*Q*cos(theta)), where theta is the angle between P and Q.
Direction of Resultant (alpha with respect to P): tan(alpha) = (Q * sin(theta)) / (P + Q * cos(theta)).
* Triangle Law of Forces: If two forces acting at a point are represented in magnitude and direction by the two sides of a triangle taken in order, their resultant is represented by the closing third side taken in the opposite order.
* Polygon Law of Forces: If a system of concurrent coplanar forces is represented in magnitude and direction by the sides of a polygon taken in order, the closing side taken in reverse order represents their resultant vector.

>> 1.3 Analytical Resolution of Coplanar Concurrent Force Systems

For complex concurrent force systems, analytical orthogonal resolution along Cartesian axes (x-y) is standard practice:
* Sum of horizontal components: Sigma Fx = F1*cos(theta1) + F2*cos(theta2) + ... + Fn*cos(thetan).
* Sum of vertical components: Sigma Fy = F1*sin(theta1) + F2*sin(theta2) + ... + Fn*sin(thetan).
* Magnitude of Resultant Vector: R = sqrt((Sigma Fx)^2 + (Sigma Fy)^2).
* Direction Angle: theta_R = atan(|Sigma Fy| / |Sigma Fx|), with quadrant placement determined by component algebraic signs.

>> 1.4 Equilibrium of Concurrent Forces & Lami’s Theorem

A particle or concurrent force system is in static equilibrium if the resultant force vanishes identically:
* Equations of Equilibrium: Sigma Fx = 0, and Sigma Fy = 0.
* Free Body Diagram (FBD): An isolated sketch of the physical body stripped of its surroundings, showing all applied active loads, gravitational weights, and reactive contact forces/cable tensions.
* Lami’s Theorem: If three coplanar concurrent forces acting at a point are in static equilibrium, each force is directly proportional to the sine of the angle between the remaining two forces.
Formula: P / sin(alpha) = Q / sin(beta) = R / sin(gamma), where alpha is angle between Q and R, beta between P and R, gamma between P and Q.

>> 1.5 Moment of a Force, Varignon’s Theorem and Couples

* Moment of a Force (M): The rotational tendency of a force about an axis or point: M_O = F * d, where d is the perpendicular moment arm from O to the force line of action. In vector notation: M_O = r x F.
* Varignon’s Theorem (Principle of Moments): The moment of a force about any point equals the algebraic sum of the moments of its constituent rectangular components about the same point: M_O(F) = M_O(Fx) + M_O(Fy).
* Force Couple: A system consisting of two equal, opposite, and non-collinear parallel forces separated by perpendicular distance d. The moment of a couple is a free vector: M = F * d. Its magnitude is invariant with respect to the reference origin.
* General Coplanar Equilibrium Equations: Sigma Fx = 0, Sigma Fy = 0, and Sigma M_O = 0 (about any arbitrary point O).

>> 1.6 Support Types and Reaction Determinations for Beams

* Roller Support: Restrains displacement perpendicular to support surface; provides 1

vertical reaction force (1 unknown).
* Hinge / Pin Support: Restrains linear displacement in both horizontal and vertical directions; provides 2 reaction forces Rx and Ry (2 unknowns).
* Fixed / Built-in Support: Restrains both linear displacements and angular rotation; provides 2 reaction forces (Rx, Ry) and 1 reactive restraint moment M (3 unknowns).

[EXAMPLE 1: Resultant of Four Coplanar Concurrent Forces]

Problem Context: Four concurrent forces act at origin O: F1 = 50 kN at 30 deg in Quadrant I; F2 = 80 kN at 45 deg in Quadrant II; F3 = 60 kN at 60 deg in Quadrant III; F4 = 40 kN directed vertically downward along negative y-axis. Determine the resultant magnitude and direction.
Solution Step 1: Resolve all forces into orthogonal components:
F1: F1x = + 50*cos(30) = + 43.30 kN, F1y = + 50*sin(30) = + 25.00 kN.
F2: F2x = - 80*cos(45) = - 56.57 kN, F2y = + 80*sin(45) = + 56.57 kN.
F3: F3x = - 60*cos(60) = - 30.00 kN, F3y = - 60*sin(60) = - 51.96 kN.
F4: F4x = 0.00 kN, F4y = - 40.00 kN.
Solution Step 2: Sum horizontal and vertical components:
Sigma Fx = 43.30 - 56.57 - 30.00 + 0 = - 43.27 kN.
Sigma Fy = 25.00 + 56.57 - 51.96 - 40.00 = - 10.39 kN.
Solution Step 3: Compute Resultant Magnitude and Direction:
R = sqrt((-43.27)^2 + (-10.39)^2) = sqrt(1872.29 + 107.95) = sqrt(1980.24) = 44.50 kN.
Direction: Both components are negative => Resultant lies in Quadrant III.
theta = atan(|-10.39| / |-43.27|) = atan(0.2401) = 13.50 deg below negative x-axis (or 193.50 deg from positive x-axis).
Conclusion: Resultant force is 44.50 kN acting at 193.50 deg counterclockwise from the positive x-axis.

[EXAMPLE 2: Equilibrium of Suspended Sphere using Lami’s Theorem]

Problem Context: A smooth uniform steel sphere of mass 100 kg (weight W = 981 N) and radius r = 200 mm is suspended by a string of length L = 300 mm tied to a smooth vertical wall. Determine: (1) Tension T in the string, (2) Normal reaction R_w exerted by the wall on the sphere.
Solution Step 1: Establish geometry and angles:
Let O be center of sphere and A be tie point on wall. In right triangle O-Wall-A:
Hypotenuse OA = string length + sphere radius = 300 + 200 = 500 mm.
Horizontal adjacent side (from center to wall contact point) = radius r = 200 mm.
cos(theta) = 200 / 500 = 0.40 => theta = 66.42 deg (angle of string with horizontal), or angle with vertical phi = 90 - 66.42 = 23.58 deg.
Solution Step 2: Free Body Diagram of the sphere:

Three concurrent forces act at sphere center O:
- Weight W = 981 N acting vertically downward (270 deg).
- Normal reaction R_w acting horizontally to the right (0 deg) from the smooth wall.
- Cable tension T pulling up and left at angle phi = 23.58 deg to the vertical.
Solution Step 3: Compute angles between forces for Lami's Theorem:
Angle between R_w and W = 90 deg.
Angle between T and W = 180 - 23.58 = 156.42 deg.
Angle between T and R_w = 90 + 23.58 = 113.58 deg.
Solution Step 4: Apply Lami's Theorem:
T / sin(90) = R_w / sin(156.42) = W / sin(113.58).
T = 981 * (sin(90) / sin(113.58)) = 981 / 0.9165 = 1070.38 N.
R_w = 981 * (sin(156.42) / sin(113.58)) = 981 * (0.4000 / 0.9165) = 428.15 N.
Conclusion: Cable tension T = 1070.38 N, Wall reaction R_w = 428.15 N.

[EXAMPLE 3: Varignon's Theorem & Non-Concurrent Resultant Location]

Problem Context: A horizontal steel bracket ABCD is subjected to three vertical forces: 20 kN downward at x = 0 m (A); 40 kN upward at x = 2 m (B); 30 kN downward at x = 5 m (D); and a concentrated clockwise couple of 50 kN-m at point C (x = 3.5 m). Determine the magnitude, nature, and position of the single resultant force measured from point A.
Solution Step 1: Calculate Resultant Force Magnitude:
Since all forces are purely vertical: R = Sigma Fy = - 20 + 40 - 30 = - 10 kN (acting downward).
Solution Step 2: Calculate Algebraic Sum of Moments about Point A:
Taking counterclockwise as positive (+), clockwise as negative (-):
M_A = [- 20 * 0] + [+ 40 * 2.0] + [- 30 * 5.0] + [- 50 (couple)]
M_A = 0 + 80 - 150 - 50 = - 120 kN-m (clockwise moment of 120 kN-m).
Solution Step 3: Apply Varignon's Theorem to locate resultant line of action:
M_A(R) = R * x_bar
Since R is downward (-10 kN), a downward force placed at distance x_bar to the right of A produces a clockwise moment about A:
- 10 * x_bar = - 120 kN-m => x_bar = 120 / 10 = 12.0 m.
Conclusion: The resultant is a single downward force of 10 kN located at a distance of 12.0 m to the right of point A.

[EXAMPLE 4: Support Reactions for a Simply Supported Beam with UDL]

Problem Context: A simply supported civil engineering beam AB of span L = 8.0 m has a pin support at A and a roller support at B. It carries a uniformly distributed load (UDL) of w = 15 kN/m over the left 4.0 m span (from x = 0 to x = 4 m), and a concentrated point load of P = 50 kN at x = 6.0 m. Calculate the support reactions at A and B.

Solution Step 1: Equivalent point load for the distributed UDL:
Total load of UDL = w * length = 15 kN/m * 4.0 m = 60 kN.
Centroid of UDL is at its midpoint: x_udl = 4.0 / 2 = 2.0 m from support A.
Solution Step 2: Apply Moment Equilibrium about support A (Sigma M_A = 0):
Clockwise moments from loads = Counterclockwise moment from roller reaction R_B:
(60 kN * 2.0 m) + (50 kN * 6.0 m) - (R_B * 8.0 m) = 0.
120 + 300 - 8.0 * R_B = 0 => 8.0 * R_B = 420 kN-m.
R_B = 420 / 8.0 = 52.50 kN (upward).
Solution Step 3: Apply Vertical Equilibrium (Sigma Fy = 0):
R_A + R_B - 60 - 50 = 0 => R_A + 52.50 = 110 kN.
R_A = 110 - 52.50 = 57.50 kN (upward).
Solution Step 4: Horizontal Equilibrium (Sigma Fx = 0):
Since all loads are vertical and roller at B offers no horizontal resistance: H_A = 0 kN.
Conclusion: Support reactions are R_A = 57.50 kN (vertical upward), R_B = 52.50 kN (vertical upward), H_A = 0 kN.

>> 1.7 University Review & Examination Questions

* Q1: State and prove Varignon’s Theorem of Moments for a system of coplanar forces.
Answer: Varignon’s Theorem states that the algebraic sum of the moments of a system of coplanar forces about any point in their plane equals the moment of their resultant about the same point. Proof: Consider concurrent forces F1, F2 acting at point P with position vector r from origin O. The resultant is R = F1 + F2. Taking the cross product of r with both sides: r x R = r x (F1 + F2) = (r x F1) + (r x F2). By definition, r x R is the moment of the resultant M_O(R), while r x F1 and r x F2 are the moments of individual forces M_O(F1) and M_O(F2). Thus, M_O(R) = Sigma M_O(F).
* Q2: Explain the physical significance of a Free Body Diagram (FBD) and enumerate the systematic steps to construct it.
Answer: A Free Body Diagram is an analytical diagram of a structural body isolated from its physical supports and adjacent elements, depicting all external active forces (applied loads, weight) and reactive forces (support reactions, cable tensions, friction). Steps: (1) Clearly identify and isolate the body of interest. (2) Replace all physical supports with equivalent reactive force/moment vectors according to their kinematic constraints. (3) Indicate all applied external loads and self-weight acting at the centre of gravity. (4) Label all unknown coordinates, inclination angles, and coordinate axes.

>> 1.8 Key Takeaways and Revision Points

* Concurrent forces meet at a common point; non-concurrent force systems create rotational moments requiring moment equilibrium.
* Lami’s theorem applies strictly to THREE concurrent coplanar forces in equilibrium.
* A couple consists of two equal and opposite parallel forces producing pure rotation

invariant of the origin point.

* Pin supports provide 2 reactions (Rx, Ry); rollers provide 1 perpendicular reaction; fixed supports provide 3 reactions (Rx, Ry, M).